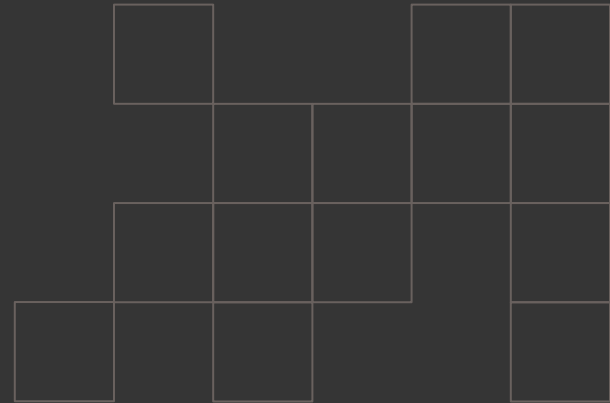


i-CES Call For Enthusiasts
2082

Intro to Machine Learning



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Introduction

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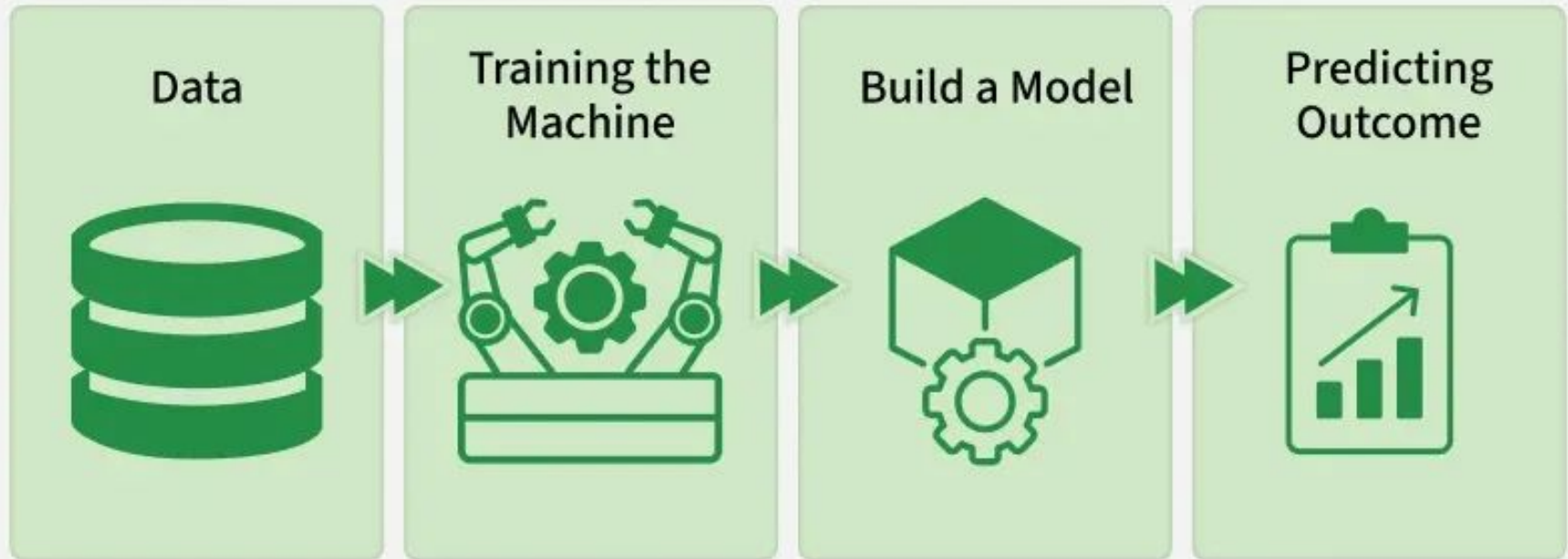
Bias and Variance

7.

Metrics (Precision, Accuracy, F1-score, etc)

Introduction

Introduction To Machine Learning



Introduction

Why Machine Learning Matters



Solves complex problems that traditional programming can't.



Handles huge amounts of data quickly.



Automates repetitive tasks with accuracy.



Improves and adapts over time with more data.

Introduction

Types of Machine Learning



Supervised Learning

Learns from labeled data.



Unsupervised Learning

Finds patterns in data without labels.



Reinforcement Learning

Learns through trial and error.

Introduction

Real-World Applications of Machine Learning



Healthcare



E-Commerce



Finance



Voice Assistants

Types of Machine Learning

1. Supervised Learning
2. Unsupervised Learning
3. Reinforcement Learning

Supervised Learning

- Supervised learning is a machine learning approach that's defined by its use of labeled data sets.

Types of Supervised Learning:

1. Regression
2. Classification

Unsupervised Learning

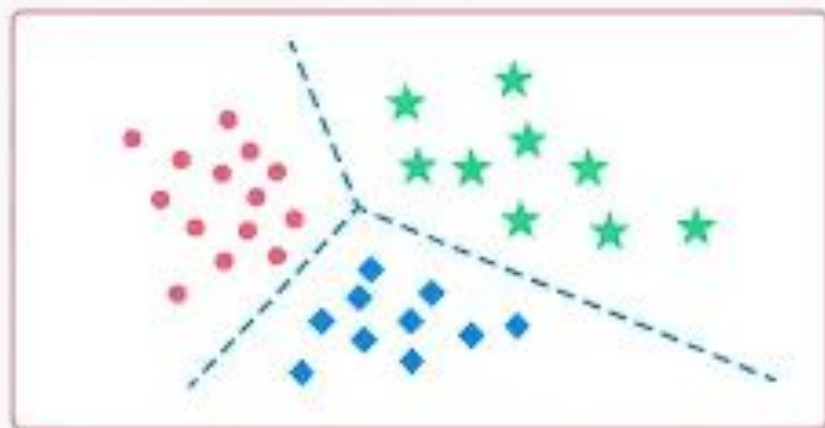
- Unsupervised learning uses machine learning algorithms to analyze and cluster unlabeled data sets.

Types of Unsupervised Learning:

1. Clustering
2. Association
3. Dimension Reduction (preprocessing)

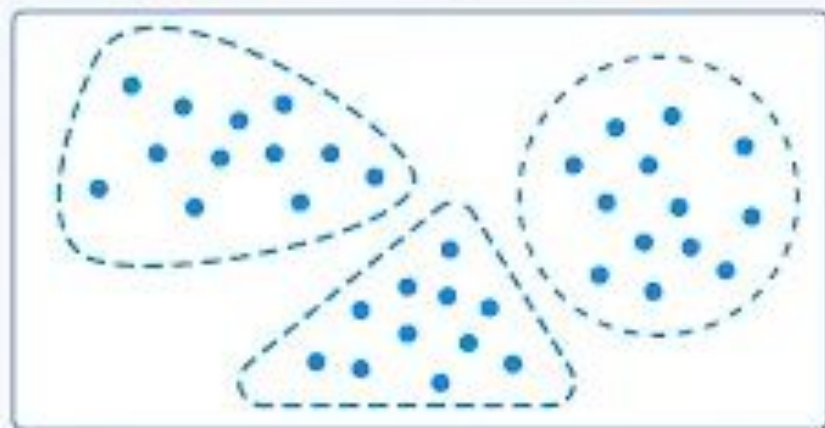
Supervised vs. Unsupervised Learning

Classification



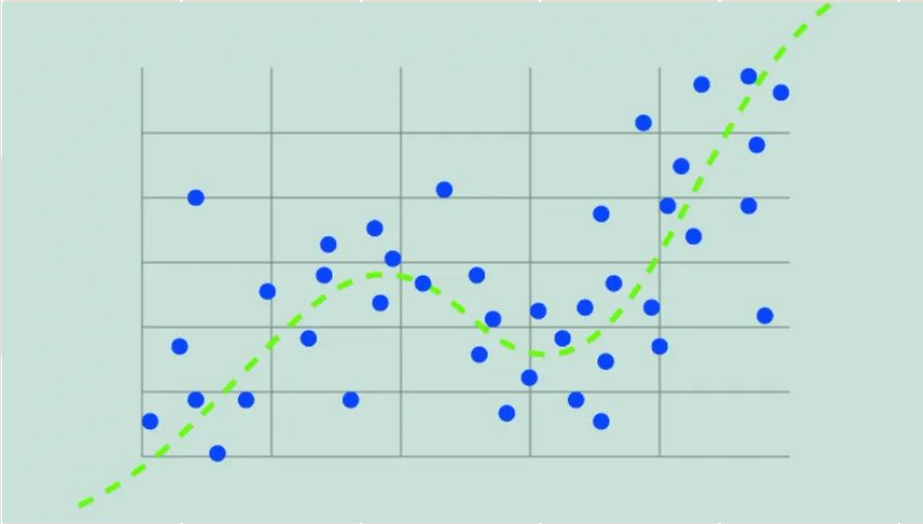
Supervised learning

Clustering



Unsupervised learning

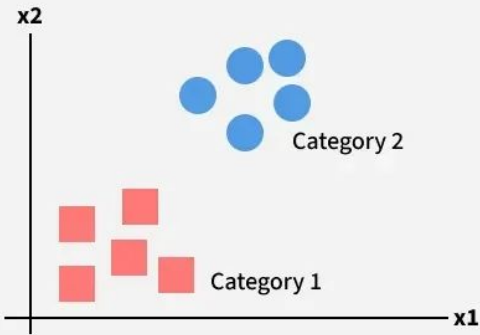
1. Regression



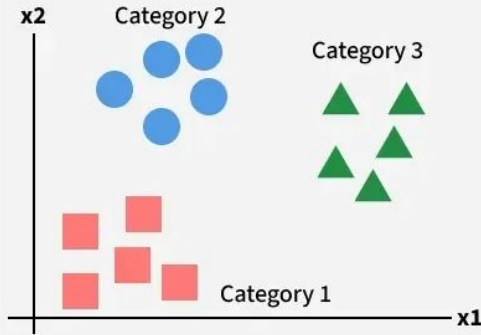
- Regression is used to understand the relationship between dependent and independent variables.
- In regression problems, the output is a continuous value, and models attempt to predict the target output.
- Regression tasks include projections for sales revenue or financial planning.

2. Classification

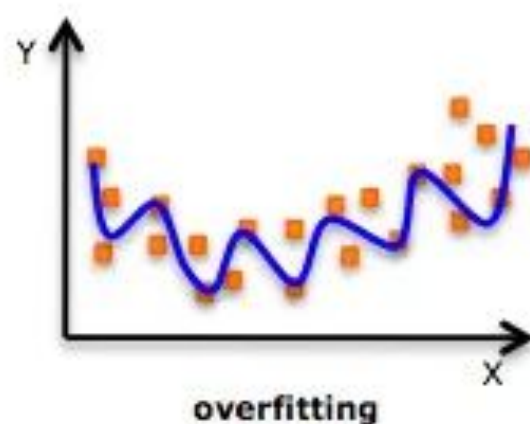
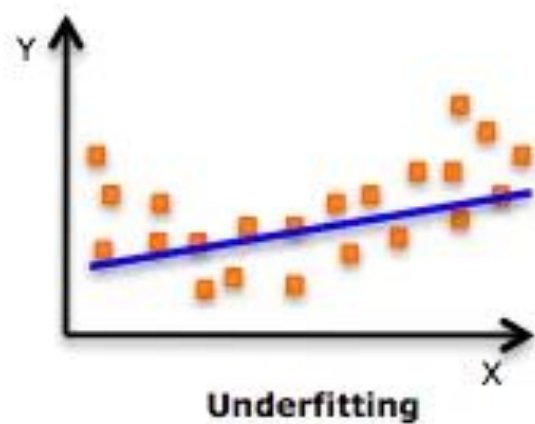
Binary Classification



Multi-class Classification



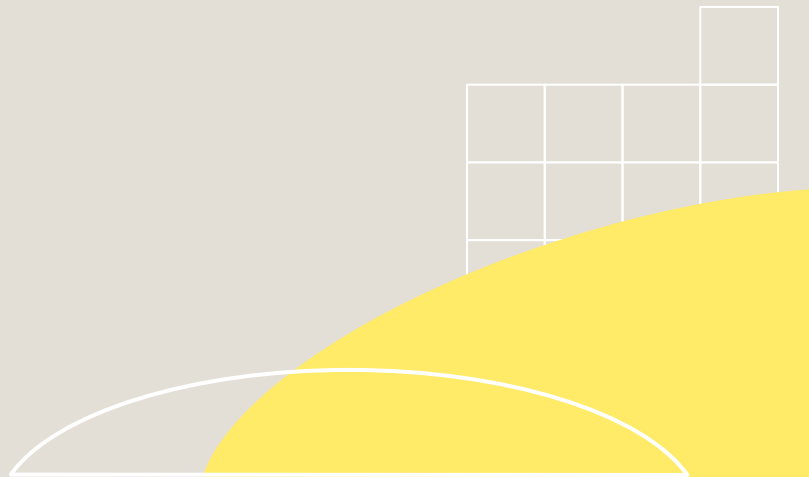
- Classification in machine learning uses an algorithm to sort data into categories.
- It recognizes specific entities within the dataset and attempts to determine how those entities should be labeled or defined.
- Real World Examples: spam mail classification, tumour classification etc.



An example of overfitting, underfitting and a model that's "just right!"

Bias and Variance

[Bias and Variance in Machine Learning - GeeksforGeeks](#)

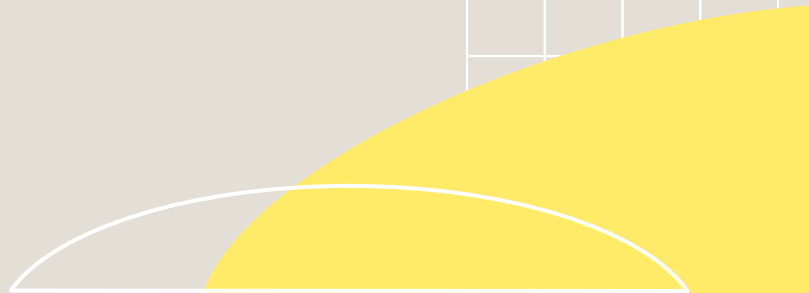
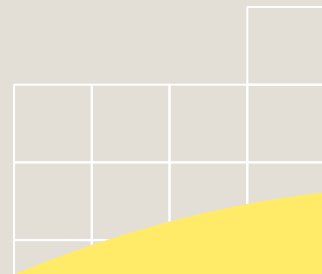


Metrics

1. Classification Metrics
 - a. Accuracy
 - b. Confusion Matrix
 - c. Precision
 - d. Recall
 - e. F1 score
2. Regression Metrics
 - a. Mean Absolute Error
 - b. Mean Squared Error
 - c. Root Mean Squared Error

Metrics CheatSheet

<https://remykarem.github.io/blog/evaluation-metrics.html>





Thank you